Information Retrieval Assignment Report-HW4

This assignment focuses on implementing the **PageRank algorithm**, one of the most important link analysis techniques in Information Retrieval. The purpose of PageRank is to determine the relative importance of web pages by analyzing the structure of links between them.

The algorithm assumes that pages receiving links from many other pages are more important. It also considers the quality of those links, meaning that links from highly ranked pages contribute more authority than links from low-ranked pages.

The PageRank formula used in this implementation is:

PR(A) = (1-d)/N + d × Σ(PR(B)/L(B))

Where:

- **PR(A)** is the PageRank score of page A
- **d** is the damping factor
- **N** is the total number of pages
- **PR(B)** is the PageRank of page B
- **L(B)** is the number of outgoing links from page B

The damping factor was set to **0.85**, which is the standard value commonly used in PageRank implementations.

Dataset Description

The dataset used for this assignment was a **10.5 MB sample** of the **WT2g (Web Track 2 Gigabyte) collection**.

WT2g is a benchmark web dataset commonly used in Information Retrieval experiments. The full dataset contains approximately 2GB of web pages and link structures. However, due to availability and computational limitations, a smaller subset was used.

The file used was:

wt2g_inlinks.txt

Each line in the dataset follows the format:

TargetPage Inlink1 Inlink2 Inlink3 ...

This structure provides the incoming links for each page and allows construction of the web graph required for PageRank computation.

Although the sample is significantly smaller than the full WT2g dataset, it preserves the graph structure necessary for validating the PageRank implementation.

Implementation

The PageRank algorithm was implemented using an iterative approach.

The implementation process consisted of the following steps:

First, the dataset was parsed to build the web graph. Each page and its corresponding incoming links were stored.

Next, all pages were initialized with equal PageRank values using:

1/N

where **N** is the total number of pages.

After initialization, sink nodes were identified. Sink nodes are pages with no outgoing links. These pages require special handling because they can absorb PageRank without redistributing it.

The PageRank values were then updated iteratively using the PageRank formula.

To determine when the algorithm had converged, **perplexity** was used as the stopping criterion.

Perplexity is calculated as:

Perplexity = $2^{(-\Sigma(PR(p) \log_2 PR(p)))}$

The algorithm was considered converged when the change in perplexity was less than 1 for four consecutive iterations.

Results

The PageRank implementation successfully ranked pages based on their importance in the link graph.

The top-ranked pages obtained from the sample dataset were:

Rank	Page ID	PageRank Score	Inlinks
1	WT21-B37-76	0.002679	2568
2	WT21-B37-75	0.001526	1704
3	WT25-B39-116	0.001470	169
4	WT23-B21-53	0.001373	198
5	WT24-B40-171	0.001245	270
6	WT23-B39-340	0.001240	274
7	WT23-B37-134	0.001205	208
8	WT08-B18-400	0.001144	1011
9	WT13-B06-284	0.001125	454
10	WT24-B26-46	0.001085	187

Additional execution statistics:

- **Dataset Size:** 10.5 MB
- **Total Pages Processed:** 183,811
- **Sink Nodes:** 66,175
- **Iterations to Converge:**118
- **Runtime:** 17.8seconds

Analysis

The results show a clear relationship between the number of incoming links and PageRank score.

The highest-ranked page, **WT21-B37-76**, also had the highest number of inlinks (2568), indicating that heavily referenced pages tend to receive greater authority.

However, the relationship is not purely based on link count.

Some pages with fewer inlinks still achieved relatively high PageRank values. This occurs because PageRank considers not only the number of incoming links but also the importance of the linking pages.

This demonstrates one of the major strengths of PageRank: it captures link quality as well as link quantity.

The convergence behavior was also consistent with expectations. The perplexity values stabilized after several iterations, confirming that the implementation was functioning correctly.

Limitations

The primary limitation of this assignment was the use of a sample dataset instead of the complete WT2g collection.

Using only a 10.5 MB subset means:

- The web graph is incomplete
- Some authoritative pages may be missing
- Rankings may differ from full-dataset results

Another limitation was the inability to implement the **HITS algorithm**.

HITS requires:

- Full document content
- Searchable document indexing
- Explicit outlink information

These resources were not available.

Conclusion

This assignment successfully implemented the PageRank algorithm and demonstrated its effectiveness in ranking web pages based on link structure.

The implementation showed that:

PageRank successfully identifies authoritative pages by analyzing incoming link patterns.

The experiment also confirmed that:

- Link quantity affects rank
- Link quality is equally important
- Perplexity provides an effective convergence criterion

Although only a sample of the WT2g dataset was used, the implementation successfully validated the correctness of the algorithm.